Schémas SQL — LUXURY CARS

Base de données PostgreSQL · Schéma complet (déjà fait & à faire) et périmètre du projet actuel

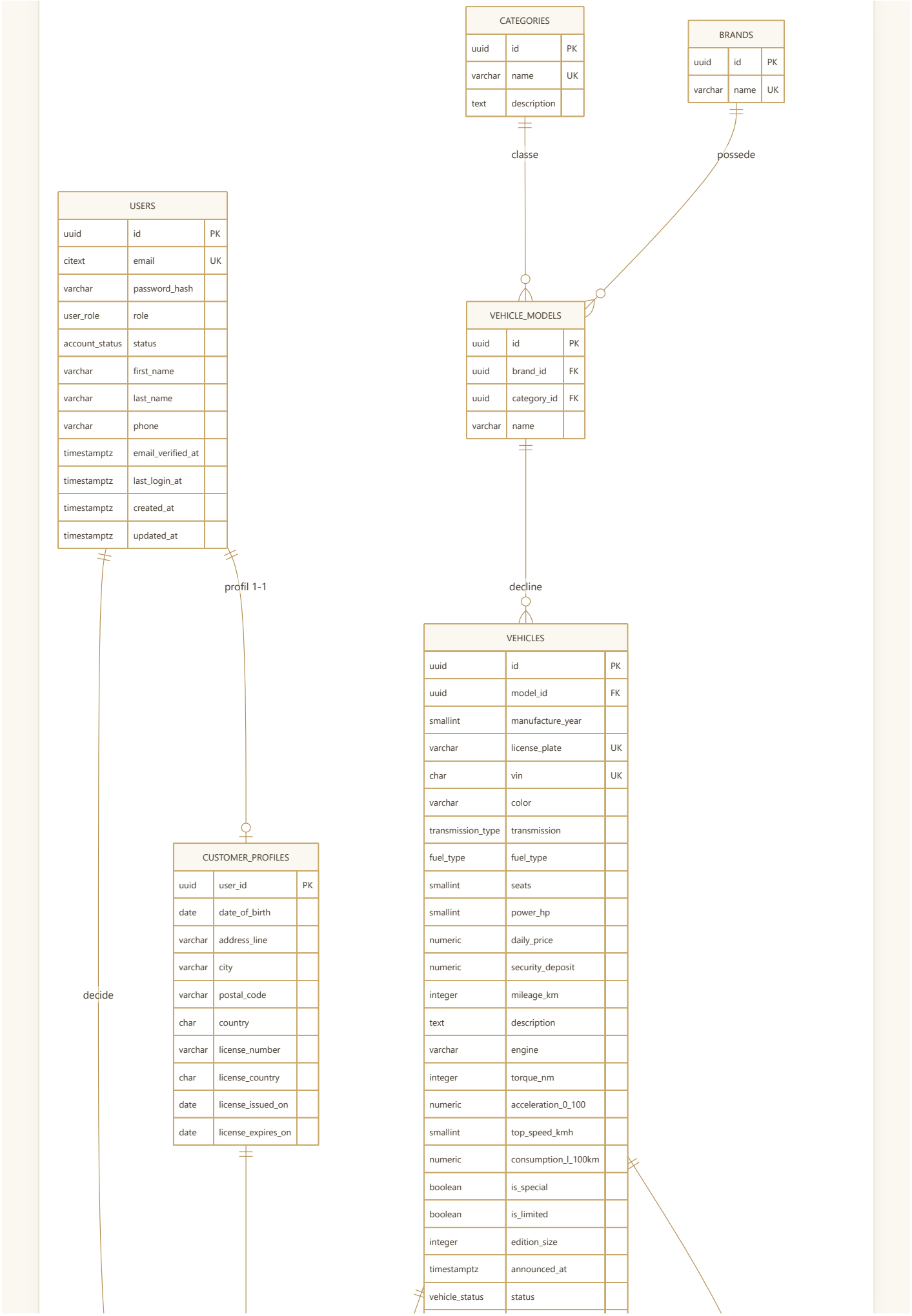
🟢 **Déjà fait** — table créée et utilisée par l'application

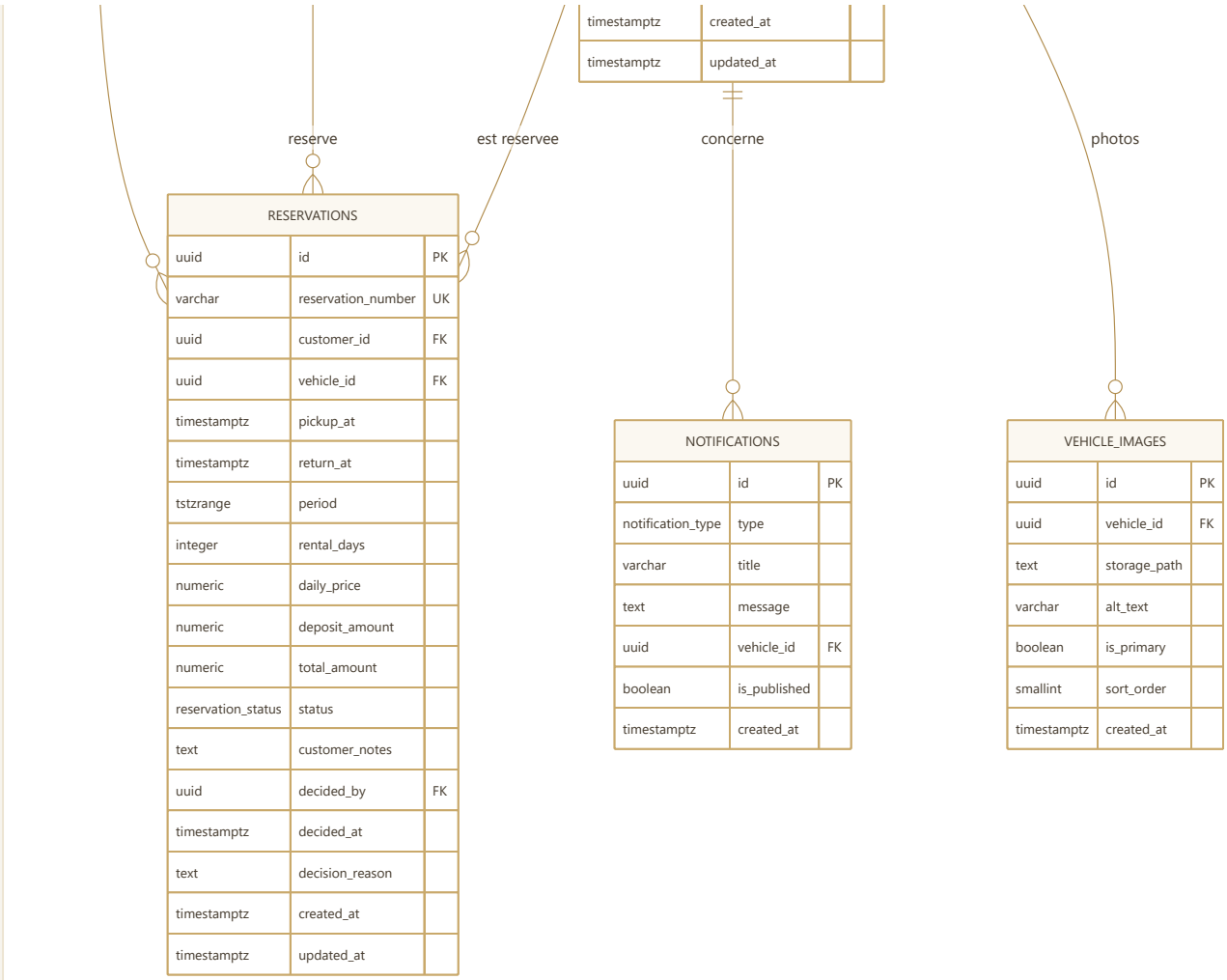
🟡 **À faire** — table créée, fonctionnalité applicative à développer

> Source de vérité : `backend/database/schema.sql` · Patch : `backend/database/patches/002-editions-and-notifications.sql` · Migrations : `backend/src/database/migrations/`
> Les **22 tables** du schéma complet existent déjà en base. « À faire » = fonctionnalité non développée, pas table manquante.

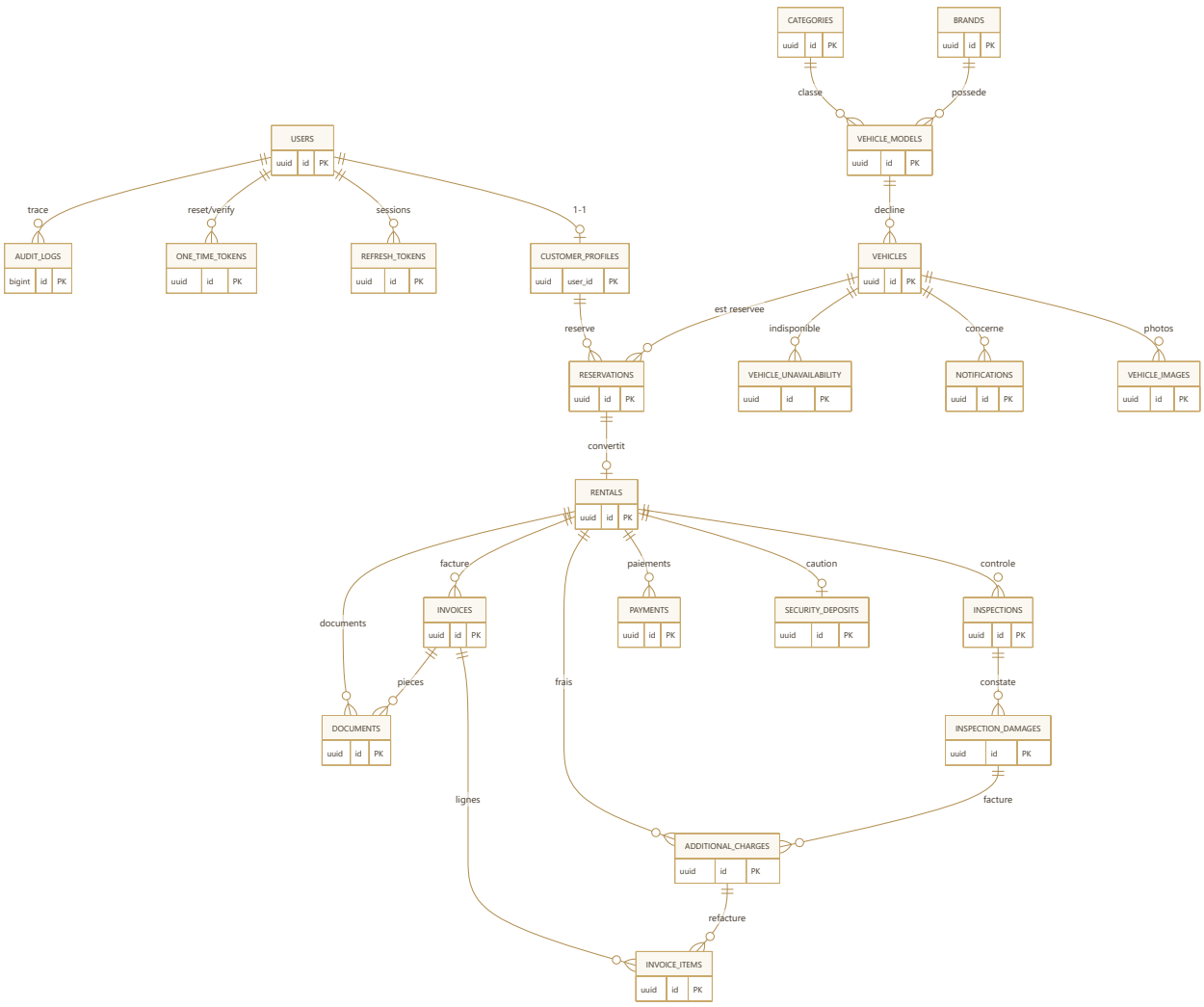
1. Périmètre du projet actuel (déjà fait) ✅

Catalogue de voitures, comptes clients, réservations et notifications.


CATEGORIES
uuid id PK
varchar name UK
text description
BRANDS
uuid id PK
varchar name UK
classe
possede
USERS
uuid id PK
citext email UK
varchar password_hash
user_role role
account_status status
varchar first_name
varchar last_name
varchar phone
timestamptz email_verified_at
timestamptz last_login_at
timestamptz created_at
timestamptz updated_at
VEHICLE_MODELS
uuid id PK
uuid brand_id FK
uuid category_id FK
varchar name
profil 1-1
decline
VEHICLES
uuid id PK
uuid model_id FK
smallint manufacture_year
varchar license_plate UK
char vin UK
varchar color
transmission_type transmission
fuel_type fuel_type
smallint seats
smallint power_hp
numeric daily_price
numeric security_deposit
integer mileage_km
text description
varchar engine
integer torque_nm
numeric acceleration_0_100
smallint top_speed_kmh
numeric consumption_l_100km
boolean is_special
boolean is_limited
integer edition_size
timestamptz announced_at
vehicle_status status
decide
CUSTOMER_PROFILES
uuid user_id PK
date date_of_birth
varchar address_line
varchar city
varchar postal_code
char country
varchar license_number
char license_country
date license_issued_on
date license_expires_on

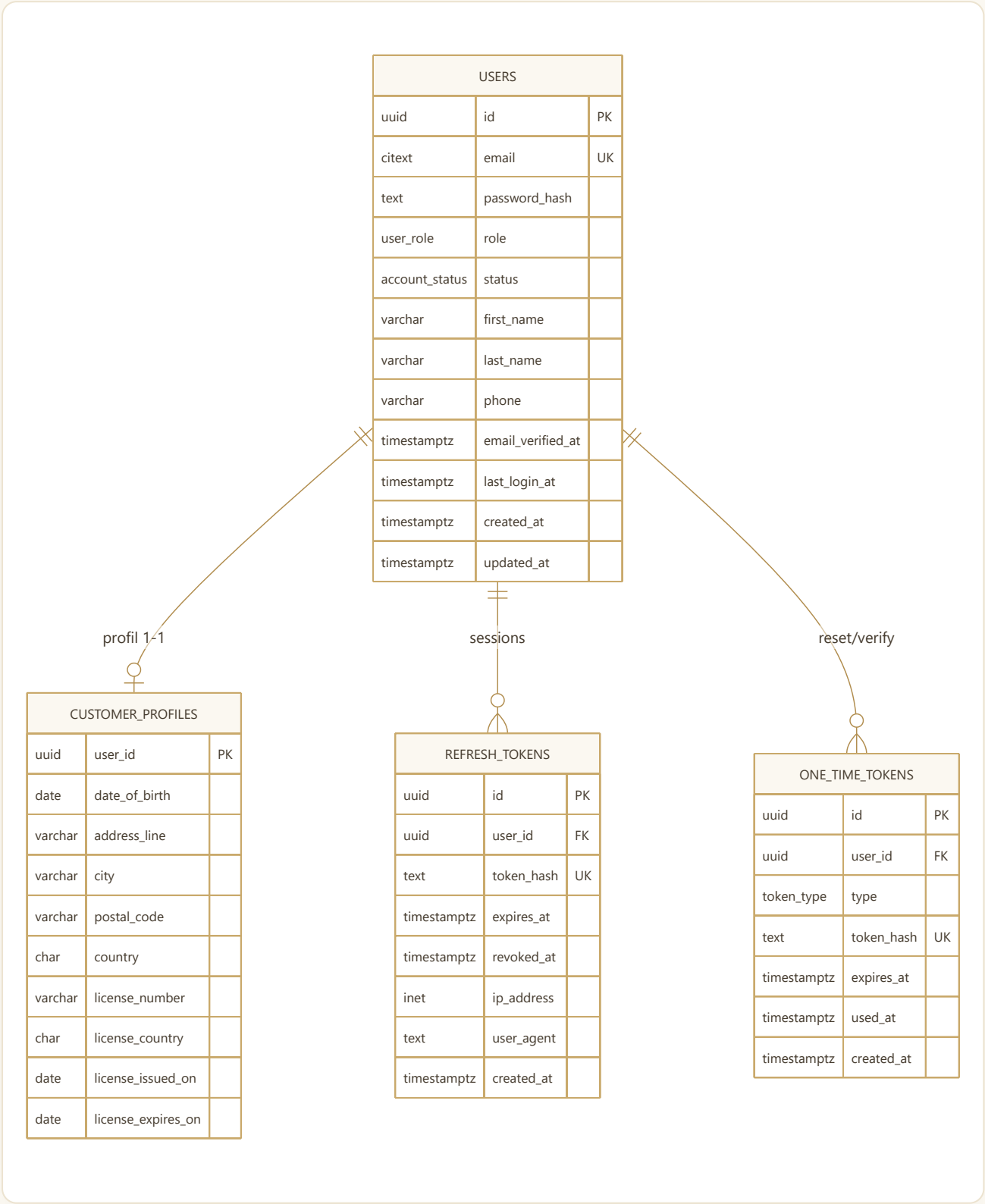

timestamptz
created_at
timestamptz
updated_at
reserve
est reservee
concerne
photos
RESERVATIONS
uuid
id
PK
varchar
reservation_number
UK
uuid
customer_id
FK
uuid
vehicle_id
FK
timestamptz
pickup_at
timestamptz
return_at
tstzrange
period
integer
rental_days
numeric
daily_price
numeric
deposit_amount
numeric
total_amount
reservation_status
status
text
customer_notes
uuid
decided_by
FK
timestamptz
decided_at
text
decision_reason
timestamptz
created_at
timestamptz
updated_at
NOTIFICATIONS
uuid
id
PK
notification_type
type
varchar
title
text
message
uuid
vehicle_id
FK
boolean
is_published
timestamptz
created_at
VEHICLE_IMAGES
uuid
id
PK
uuid
vehicle_id
FK
text
storage_path
varchar
alt_text
boolean
is_primary
smallint
sort_order
timestamptz
created_at

2. Schéma complet — déjà fait ✅ vs à faire 🟡

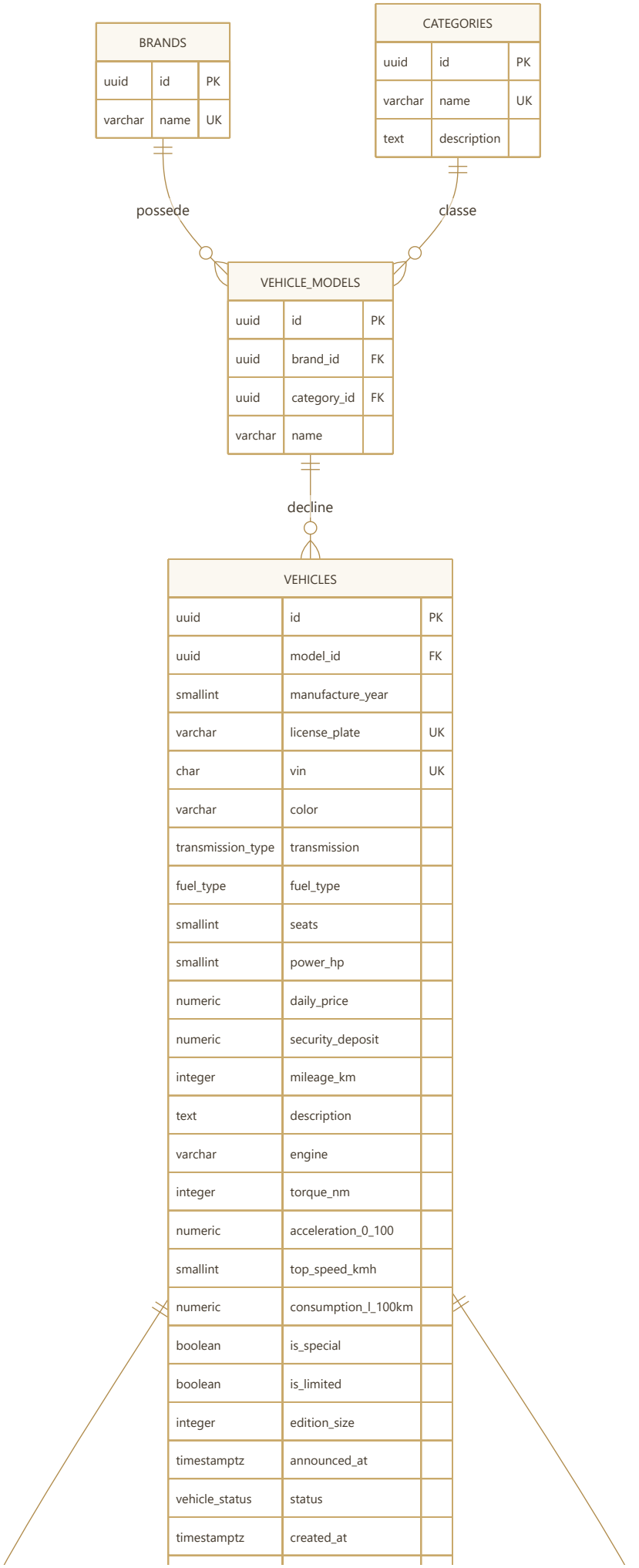


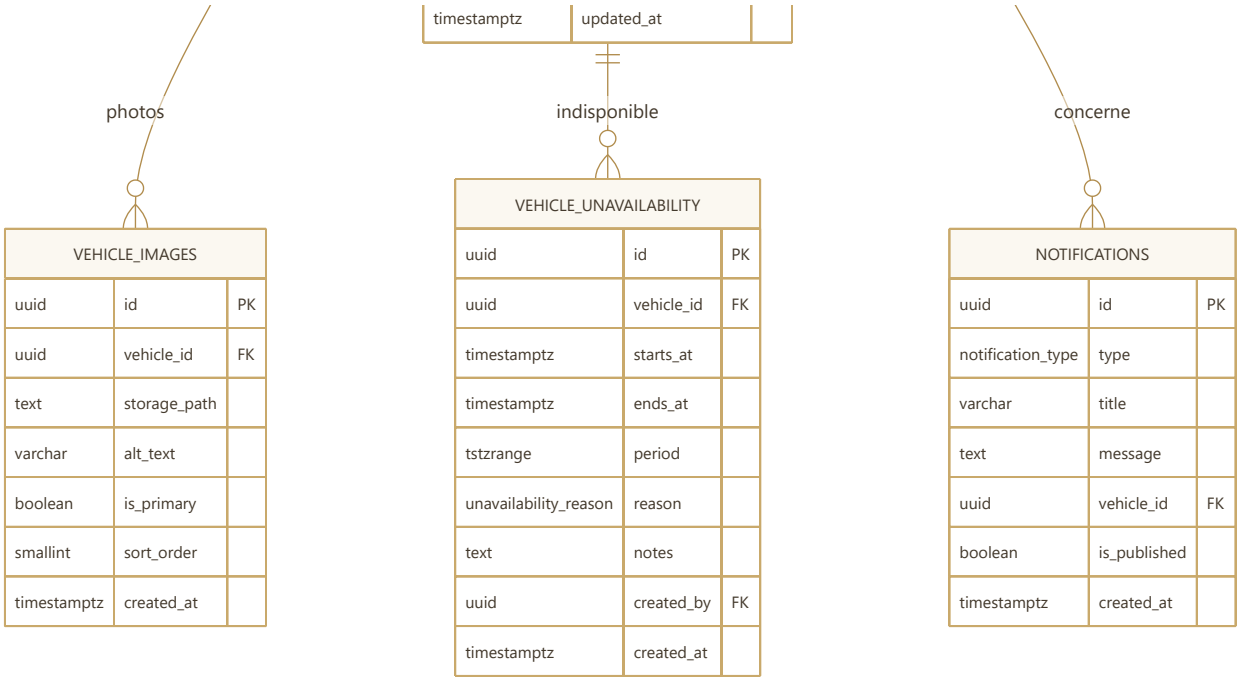
3. Détail par domaine

A. Authentification & utilisateurs

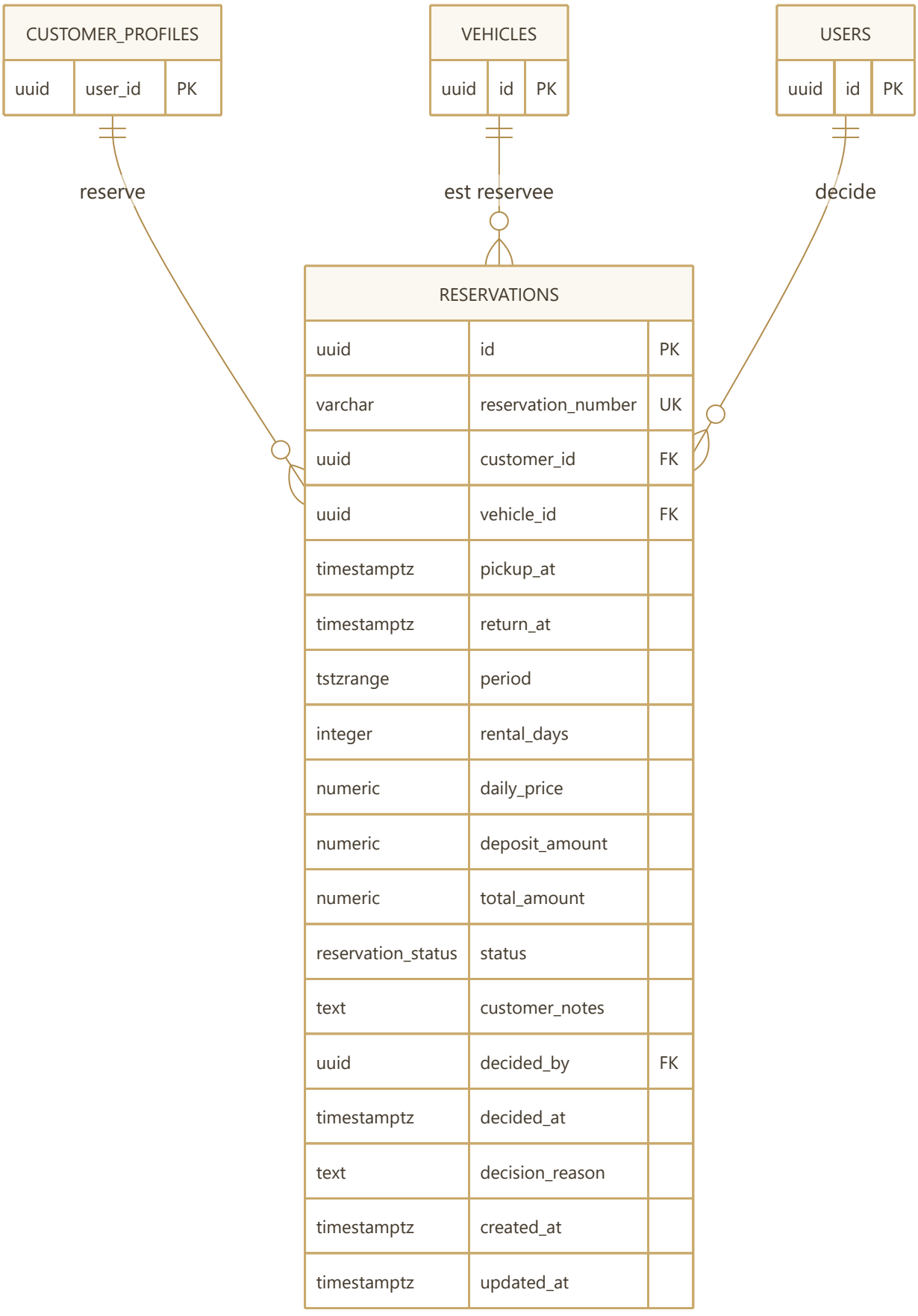


B. Catalogue de véhicules


BRANDS
uuid id PK
varchar name UK
CATEGORIES
uuid id PK
varchar name UK
text description
possede
classe
VEHICLE_MODELS
uuid id PK
uuid brand_id FK
uuid category_id FK
varchar name
decline
VEHICLES
uuid id PK
uuid model_id FK
smallint manufacture_year
varchar license_plate UK
char vin UK
varchar color
transmission_type transmission
fuel_type fuel_type
smallint seats
smallint power_hp
numeric daily_price
numeric security_deposit
integer mileage_km
text description
varchar engine
integer torque_nm
numeric acceleration_0_100
smallint top_speed_kmh
numeric consumption_l_100km
boolean is_special
boolean is_limited
integer edition_size
timestamptz announced_at
vehicle_status status
timestamptz created_at

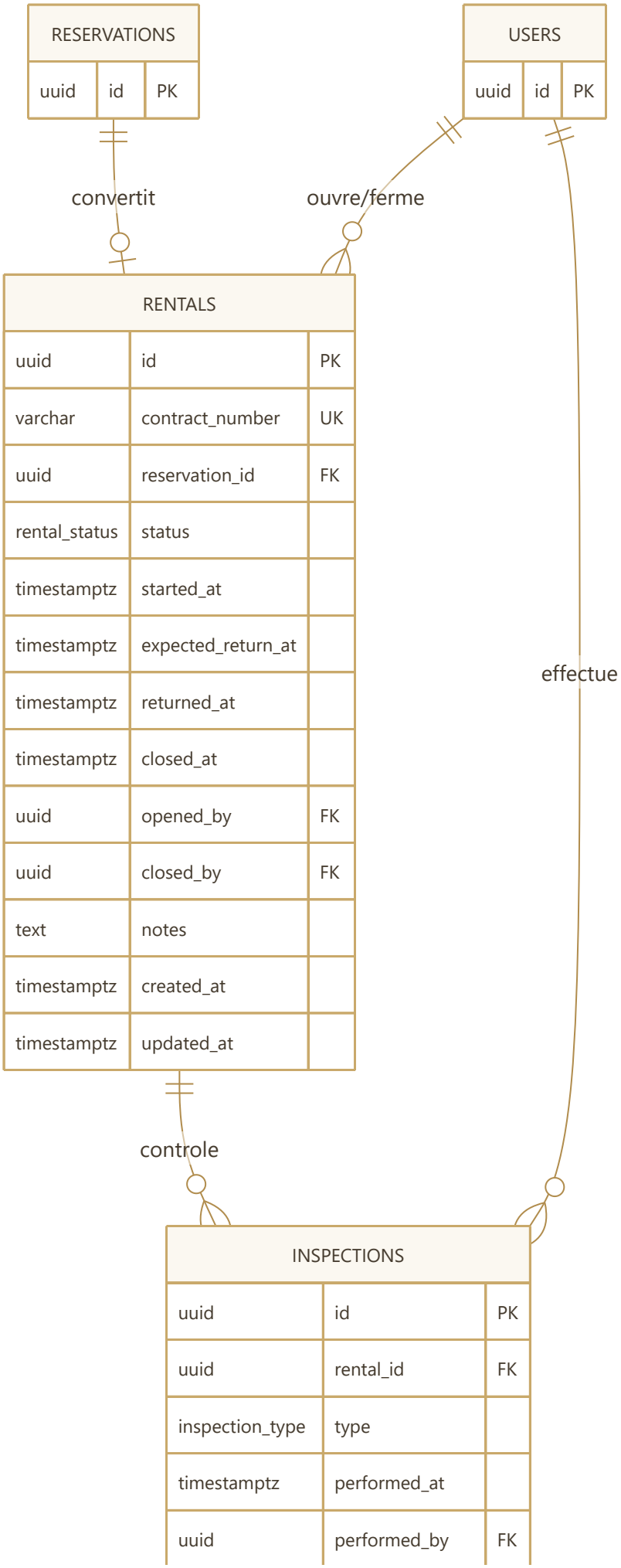

timestamptz
updated_at
photos
indisponible
concerne
VEHICLE_IMAGES
uuid
id
PK
uuid
vehicle_id
FK
text
storage_path
varchar
alt_text
boolean
is_primary
smallint
sort_order
timestamptz
created_at
VEHICLE_UNAVAILABILITY
uuid
id
PK
uuid
vehicle_id
FK
timestamptz
starts_at
timestamptz
ends_at
tstzrange
period
unavailability_reason
reason
text
notes
uuid
created_by
FK
timestamptz
created_at
NOTIFICATIONS
uuid
id
PK
notification_type
type
varchar
title
text
message
uuid
vehicle_id
FK
boolean
is_published
timestamptz
created_at

C. Réservations


CUSTOMER_PROFILES
uuid
user_id
PK
VEHICLES
uuid
id
PK
USERS
uuid
id
PK
reserve
est reservee
decide
RESERVATIONS
uuid
id
PK
varchar
reservation_number
UK
uuid
customer_id
FK
uuid
vehicle_id
FK
timestamptz
pickup_at
timestamptz
return_at
tstzrange
period
integer
rental_days
numeric
daily_price
numeric
deposit_amount
numeric
total_amount
reservation_status
status
text
customer_notes
uuid
decided_by
FK
timestamptz
decided_at
text
decision_reason
timestamptz
created_at
timestamptz
updated_at

> Contrainte `EXCLUDE USING gist` : le double booking est interdit (chevauchement de période pour une même voiture, statuts pending/confirmed/converted).

D. Locations, contrôles et retour

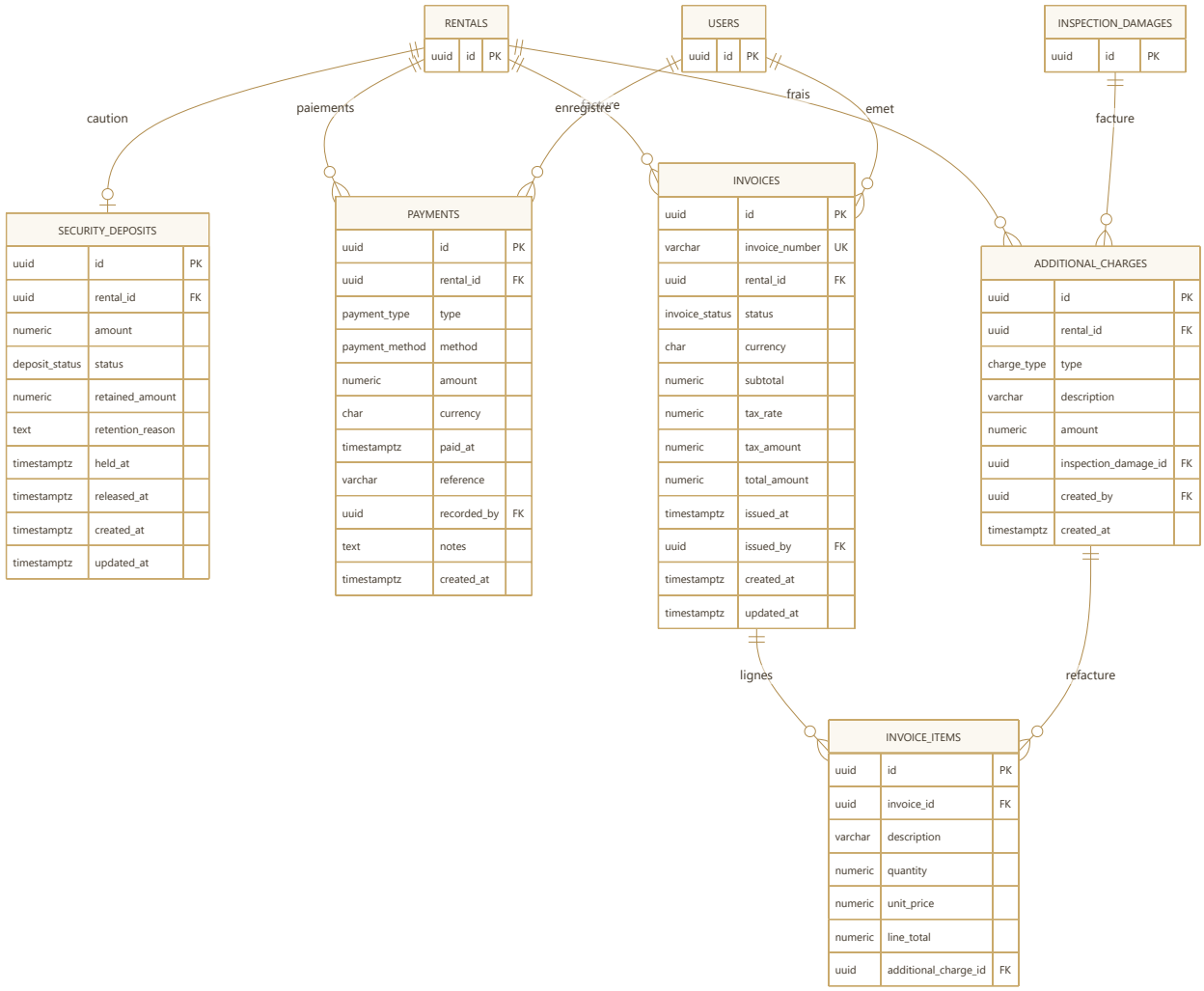

RESERVATIONS
uuid
id
PK
USERS
uuid
id
PK
convertit
ouvre/ferme
RENTALS
uuid
id
PK
varchar
contract_number
UK
uuid
reservation_id
FK
rental_status
status
timestamptz
started_at
timestamptz
expected_return_at
timestamptz
returned_at
timestamptz
closed_at
uuid
opened_by
FK
uuid
closed_by
FK
text
notes
timestamptz
created_at
timestamptz
updated_at
effectue
controle
INSPECTIONS
uuid
id
PK
uuid
rental_id
FK
inspection_type
type
timestamptz
performed_at
uuid
performed_by
FK

integer	mileage_km	
smallint	fuel_level_pct	
condition_rating	overall_condition	
text	notes	

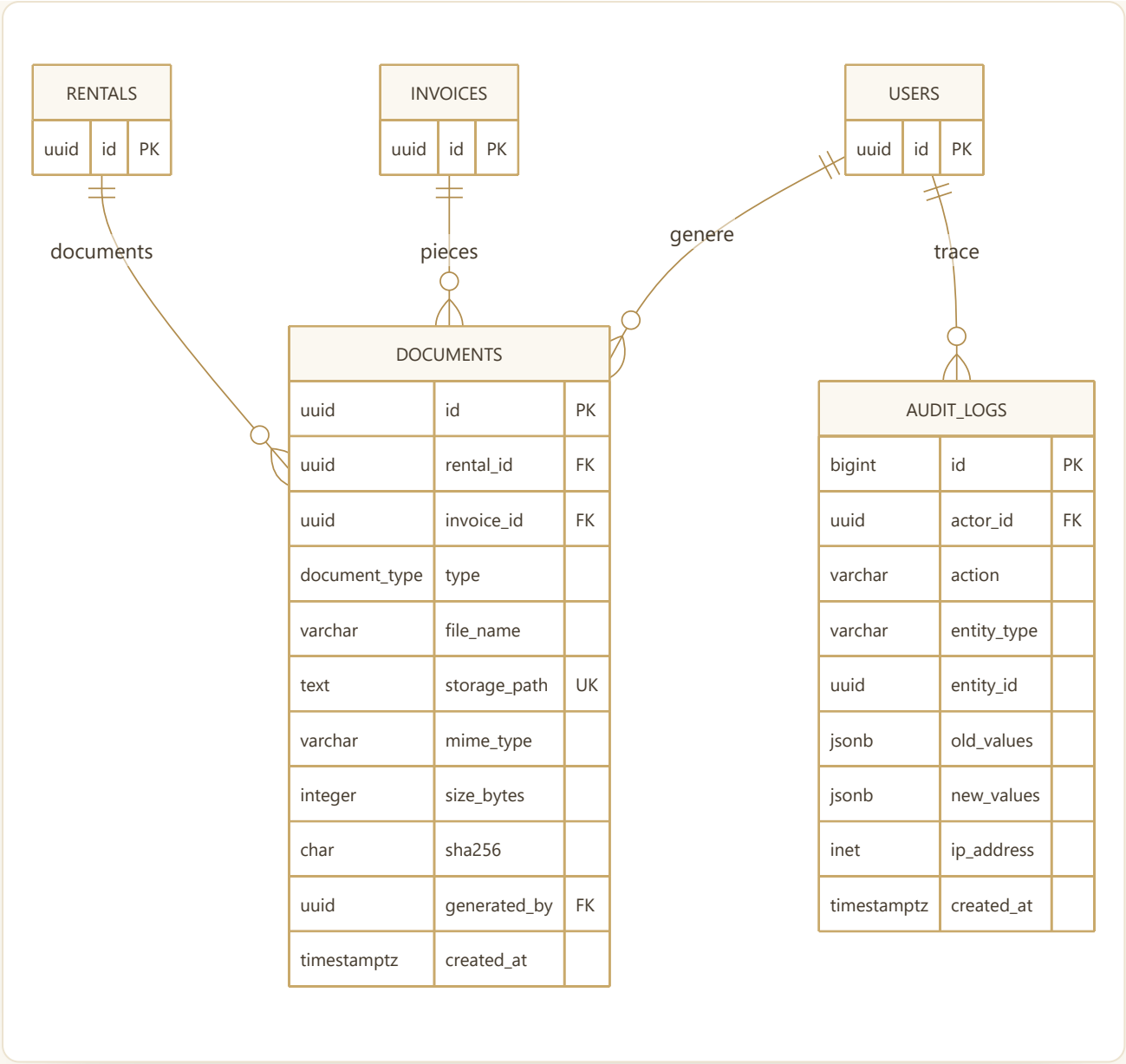


INSPECTION_DAMAGES		
uuid	id	PK
uuid	inspection_id	FK
varchar	vehicle_zone	
text	description	
damage_severity	severity	
numeric	estimated_repair_cost	
text	photo_path	

E. Argent : cautions, paiements, frais, factures



F. Documents & journal d'audit


RENTALS
uuid
id
PK
INVOICES
uuid
id
PK
USERS
uuid
id
PK
documents
pieces
genere
trace
DOCUMENTS
uuid
id
PK
uuid
rental_id
FK
uuid
invoice_id
FK
document_type
type
varchar
file_name
text
storage_path
UK
varchar
mime_type
integer
size_bytes
char
sha256
uuid
generated_by
FK
timestamptz
created_at
AUDIT_LOGS
bigint
id
PK
uuid
actor_id
FK
varchar
action
varchar
entity_type
uuid
entity_id
jsonb
old_values
jsonb
new_values
inet
ip_address
timestamptz
created_at

4. Énumérations PostgreSQL

Enum	Valeurs
`user_role`	customer, staff, admin
`account_status`	invited, active, suspended
`token_type`	password_reset, email_verification
`transmission_type`	automatic, manual, semi_automatic
`fuel_type`	petrol, diesel, hybrid, plug_in_hybrid, electric
`vehicle_status`	active, maintenance, retired
`unavailability_reason`	maintenance, repair, inspection, other
`reservation_status`	pending, confirmed, rejected, cancelled, expired, converted
`rental_status`	active, returned, closed
`inspection_type`	handover, return
`condition_rating`	excellent, good, fair, damaged
`damage_severity`	minor, moderate, major
`payment_type`	rental_payment, deposit_payment, deposit_refund, charge_payment, refund
`payment_method`	cash, card, bank_transfer
`deposit_status`	pending, held, released, partially_retained, retained
`charge_type`	late_return, fuel, mileage, damage, cleaning, other
`invoice_status`	draft, issued, paid, void
`document_type`	contract, invoice, inspection_report, other
`notification_type`	new_vehicle, special_edition, limited_edition, general

5. Vues, fonctions & triggers

Objet	Type	Rôle
`find_available_vehicles(p_from, p_to)`	Fonction SQL	Véhicules réservables sur une période

Objet	Type	Rôle
`v_dashboard_summary`	Vue	Compteurs admin (véhicules, réservations, locations, clients)
`v_monthly_revenue`	Vue	Chiffre d'affaires mensuel (cautions exclues)
`set_updated_at()`	Trigger function	Met à jour `updated_at`
`trg_reservation_vs_unavailability()`	Trigger	Bloque réservation vs indisponibilité
`trg_unavailability_vs_reservation()`	Trigger	Bloque indisponibilité vs réservation
`trg_inspection_mileage()`	Trigger	Kilométrage retour ≥ départ

6. Récapitulatif déjà fait / à faire

#	Table	Domaine	État
1	`users`	Auth	✅ Déjà fait
2	`customer_profiles`	Auth	✅ Déjà fait
3	`refresh_tokens`	Auth	🟡 À faire
4	`one_time_tokens`	Auth	🟡 À faire
5	`brands`	Catalogue	✅ Déjà fait
6	`categories`	Catalogue	✅ Déjà fait
7	`vehicle_models`	Catalogue	✅ Déjà fait
8	`vehicles`	Catalogue	✅ Déjà fait
9	`vehicle_images`	Catalogue	✅ Déjà fait (stockage manuel)
10	`vehicle_unavailability`	Catalogue	🟡 À faire
11	`notifications`	Catalogue	✅ Déjà fait
12	`reservations`	Réservations	✅ Déjà fait
13	`rentals`	Location	🟡 À faire
14	`inspections`	Location	🟡 À faire
15	`inspection_damages`	Location	🟡 À faire
16	`security_deposits`	Argent	🟡 À faire
17	`payments`	Argent	🟡 À faire

#	Table	Domaine	État
18	`additional_charges`	Argent	🟡 À faire
19	`invoices`	Argent	🟡 À faire
20	`invoice_items`	Argent	🟡 À faire
21	`documents`	Documents	🟡 À faire
22	`audit_logs`	Audit	🟡 À faire
